INCH-POUND

MIL-DTL-53126A
16 October 2012
SUPERSEDING
MIL-E-53126(ME)
20 April 1992

DETAIL SPECIFICATION

POWER DISTRIBUTION, ILLUMINATION SYSTEMS, ELECTRICAL

GENERAL SPECIFICATION FOR

This specification is approved for use within US Army Research Development and Engineering Command, Communications and Electronics Research and Development Center, Product Realization Directorate, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers soldier-portable, power distribution systems for use with the DoD Standard Family of Generator Sets and illumination utility sets. Three of the end items are three phase systems, one is a single phase system, and the utility set contains extension cords, branch circuits and 1ighting equipment. The three phase systems can be used in various combinations with each other to create a power grid which is generally 1imited to a 100 feet radius per system (i.e. 300 feet total) in order to prevent excessive voltage loss. The utility set can be used with any of the systems to achieve internal shelter wiring and lighting.

1.2 Classification. The Power Distribution Illumination System, Electrical (herein after to be known as PDISE) are of the following types, as specified herein (see 6.2).

1.2.1 Type. The PDISE systems will be of the following types:

Type I	Feeder System, electrical, three phase, single source
Type II	Feeder System, electrical, three phase, multiple sources
Type III	Distribution System, electrical, three phase
Type IV	Distribution System, electrical, single phase
Type V	Illumination System, single phase
Type VI	Universal Adapter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army RDECOM CERDEC PRD, ATTN: RDER-PRA-GS, Fort Belvoir, VA 22060-5817 Or e-mail: daniel.ojeifoh@us.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at: https://assist.dla.mil

AMSC N/A FSC 6150

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1.3 Rating. The PDISE are to be rated as follows:

Rating	Model	Description
200 Amps/Phase (A/P)	M200 A/P	Feeder System, electrical, three phase
100 Amps/Phase (A/P)	M100 A/P	Feeder System, electrical, three phase
60 Amps/Phase (A/P)	M60 A/P	Distribution System, electrical, single phase
40 Amps/Phase (A/P)	M40 A/P	Distribution System, electrical, three phase

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of the specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SPECIFICATIONS

FEDERAL

A-A-1898	Cushioning Material, Celulosic, Packaging
A-A-3174	Plastic Sheet, Polyolefin

DEPARTMENT OF DEFENSE

MIL-DTL-52286	Cable Assembly, Power, Electrical (with Molded On Terminations), General Specifications for
MIL-DTL-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
SAE-AS34011	Connectors, Receptacle, Electric, Cable Connecting, Front Release, Crimp Contact, AN Type
SAE-AS34021	Connector, Receptacle, Electric, Box Mounting, Front Release, Crimp Contact, AN Type
SAE-AS34061	Connectors, Plug, Electric, Front Release, Crimp Contact, AN Type
MS90555	Connector, Receptacle, Electrical, Wall Mounting Class L (Power Source Receptacle)

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MS90556	Connector, Plug, Electrical, Straight, Class L
MS90557	Connector, Receptacle, Electrical, Cable Connecting (without Coupling Ring), Class L
MS90558	Connector, Plug, Electrical Wall Mounting, (with Coupling Ring) Class L (Equipment)

MILITARY

MIL-DTL-22992	Connectors, Plugs and Receptacles, Electrical, Waterproof, Quick Disconnect, Heavy Duty Tape, General Specification for
MIL-DTL-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-DTL-53072	Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection
MIL-DTL-64159	Camouflage Coating. Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant
MIL-L-85762	Lighting, Aircraft, Interior, Night Vision Imaging System (NVIS) Compatible
MIL-PRF-121	Barrier Materials, Greaseproof, Waterproof, Flexible, Heat-Sealable
MIL-T-704	Treatment and Painting of Material

(See Supplement 1 for a list of associated specification sheets).

STANDARDS

FEDERAL

FED-STD-595-34094	Color Used in Government Procurement

DEPARTMENT OF DEFENSE

MIL-STD-129	Military Marking for Shipment and Storage
MIL-STD-209	Lifting and Tiedown Provisions
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	System Safety
MIL-STD-889	Dissimilar Metals
MIL-STD-1472	Human Engineering
MIL-STD-1916	DoD Preferred Methods for Acceptance of Product

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MIL-STD-2073-1E Standard Practice for Military Packaging

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 1911-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

2.2.2.1 Documents.

TM 9-6150-226-13 Technical Manual Operator and Field Maintenance Manual for Distribution Illumination Systems, Electrical (DISE) and Power Distribution Illumination Systems, Electrical (PDISE)

(Copies of the technical manuals required in connection with specific acquisition functions should be obtained from the US Army Communications and Electronics Command, ATTN: AMSEL-LCL-ECM, Aberdeen Proving Ground, MD 21005-1846.)

2.2.2.2 Drawings.

TABLE I – Drawing list

DRAWING#	NOMENCLATURE	DRAWING #	NOMENCLATURE
CECOM & PM-MEP Drawings - Cage Code: 97403 and 30554			
13230E7022	Plate, Identification	13229E6352	Distribution System, Electrical, Single Phase, 60 Amperes/Phase
13229E6353	Distribution System, Electrical, Single Phase, 40 Amperes/Phase	13226E7046	Utility Assembly, Electrical, Receptacles and Lighting
13230E7021	Packaging, PDISE M40/M60 A/P	08-26001	Packaging, PDISE M60
08-26000	Packaging, PDISE M40	13229E6351	Feeder System, Electrical, Three Phase, 100 Amperes/Phase
13230E7020	Packaging, PDISE M100 A/P	08-26002	Packaging, PDISE M100
08-26003	Packaging, PDISE M200	13229E6350	Feeder System, Electrical, Three Phase, 200 Amperes/Phase
13229E6362	Container, Storage, Utility		

2.2.2.3 Catalogs and manuals.

FM 3-11.4 Multiservice Tactics, Techniques, and Procedures for

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Nuclear, Biological and Chemical (NBC) Protection

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

ASTM D1974/D1974M	Standard Practice for Methods of Closing, Sealing, and Reinforcing Fiberboard Boxes
ASTM D3953	Steel and Seals, Flat, Strapping
ASTM D4675	Materials, Flat Strapping, Selection and Use of
ASTM D5118/D5118	Standard Practice for Fabrication of Fiberboard Shipping Boxes
ASTM D5486/D65486M	Standard Specification for Pressure-Sensitive Tape for Packaging, box Closure, and Sealing
ASTM D6251/D625M	Natural Wood-Cleaterd Panelboard Shipping Boxes

(Application for copies should be addressed to the 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or copies are available from http://www.astm.org)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025	General Requirements for the Competence of Testing and Calibration Laboratories – Second Edition
ISO 10012	Quality Assurance Requirements for Measuring Equipment

(Application for copies should be addressed to the ISO Copyright office, Case postale 56, CH-1211 Geneva 20 or copies are available from www.iso.org)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA WD 6	Wiring, Devices – Dimensional Requirements

(Applications for copies should be addressed to the National Electrical Manufacturers Association, 1300 N 17th Street, Suite 1752, Rosslyn, VA 22209 or copies are available from www.nema.org)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC

ANSI Z535.2	Environmental and Facility Safety Signs

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(Applications for copies should be addressed to the American National Institute, Inc. 1899 L Street, NW 11th Floor, Washington, DC 20036 or copies are available from www.ansi.org)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	National Electric Code (NEC)

(Application for copies should be addressed to the National Fire Protection Association International, 1 Batterymarch Park, Quincy, MA 02269 or copies are available from www.nfpa.org)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-ARP1199	Devices, Electric Overcurrent Protective, Selection, Application, and Inspection of
SAE AS34011	Connectors, Receptacle, Electric, Cable Connecting, Front Release, Crimp Contact, AN Type
SAE AS34021	Connector, Receptacle, Electric, Box Mounting, Front Release, Crimp Contact, AN Type with
SAE AS34061	Connectors, Plug, Electric, Front Release, Crimp Contact, AN Type
SAE AS50151	Connectors, Electrical, Circular Threaded, AN Type, General Specification For

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001 USA or copies are available from www.sae.org)

UNDERWRITER'S LABORATORIES INC. (UL)

UL 943	Interrupter, Ground-Fault Circuit - Standard For Safety

(Application for copies should be addressed to Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062-2096 or copies are available from www.ul.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and IAW the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern.

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3.1.1 Drawings. The drawings forming a part of this specification (as listed in the applicable specification sheets, MIL-DTL-53126/1, /2, /3, /4, /5), are end product drawings. No deviation from the prescribed drawing requirements is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the set. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection IAW 4.1.

3.3 Materials. All material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be of sufficient durability to meet the performance requirements, in any of the environments, specified herein. The PDISE and accessories shall not be made from, leach, or otherwise develop toxins, hazardous materials, or ozone depleting substances.

3.3.1 Deterioration prevention and control. The PDISE and accessories shall be fabricated from compatible materials, inherently corrosion resistant, or treated to provide protection against the various forms of corrosion and deterioration to which they are susceptible.

3.3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life-cycle costs. Used, rebuilt, or remanufactured components, pieces, and parts shall not be used.

3.3.3 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other unless protected against galvanic corrosion. Methods of protection against galvanic corrosion shall be IAW MIL-STD-889.

3.3.4 Design. The PDISE design shall be as specified herein and IAW the applicable specification sheets. The PDISE shall be capable of interfacing with all members of the DoD Standard Family of Generator sets which are capable of providing 120/208 V, 3 phase or 120 V, single phase power.

3.3.5 PDISE inputs. The PDISE shall be capable of connecting one generator set (Type I PDISE) or multiple generator sets (Type II PDISE). The PDISE input shall interface with the generator set using a 5-wire system. The 5-wire system shall consist of individual conductors per phase, neutral and ground/bonding. The single phase distribution system (Type III PDISE) shall use a 3-wire system where the phase, neutral and ground/bonding connections are individual conductors.

3.3.6 Main AC circuit interrupter. Each PDISE shall have a main AC circuit interrupter rated for the amperage capability of the system. This circuit interrupter shall be capable of disconnecting all input power from the output circuits. The AC circuit interrupter shall provide overload protection for each PDISE. The AC circuit interrupter shall activate at 125% rated current within 880 seconds. At loads above 125%, the AC circuit interrupter shall activate using an inverse time principle.

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3.3.7 Output circuits. Each PDISE shall have output circuits as specified in the specification sheets. Each circuit shall be capable of being isolated from the main circuit by the operator. Output circuits shall meet the requirements of the NPFA 70.

3.3.8 Overload protection. Each output circuit shall be protected from current overload. The overload protection shall be accomplished by the use of a resettable circuit breaker rated for 50/60 Hz operation, with solderless connector terminals, IAW the requirements of SAE ARP1199. The overload protection shall not trip for currents up to 125% of rated current for the circuit. This device shall activate within 880 seconds when 125% of rated current in any phase is reached. At currents above 125%, the AC circuit interrupter shall activate using an inverse time principle. Circuit breakers shall be capable of accurate operation in temperatures ranging from -40 °F to 185 °F.

3.3.8.1 Control panel. All controls and indicators shall be mounted on a control panel that is accessible from the exterior of the PDISE without the use of tools. The controls shall be protected from accidental activation/deactivation. Indicators shall be used to indicate status of the input and output circuits of the PDISE. The indicators and controls shall be separately grouped to indicate input(s) and outputs. The output circuit controls shall identify which circuit they are associated with. Circuit breakers shall be installed so that they are all aligned, in the "on" and "off" positions; and ganged for three phase circuits when appropriate. The PDISE control panel shall be protected from the environment by a hinged cover.

3.3.9 Housing. The PDISE shall be housed in a suitable container to prevent damage due to transportability, and operational environment requirements specified herein. The PDISE shall be skid mounted where the bottom of the housing shall be no less than 7 inches from the ground. The housing shall include:

a. A permanently attached hinged cover with a means to close the unattended cover, and latching device(s). A wiring diagram, with each circuit labeled, shall be permanently attached to the inside of the cover.

b. Carrying handles, capable of supporting the unit indefinitely, suitable for use by personnel wearing arctic mittens. The placement and quantity of handles shall be consistent with an individual lift capability of 40 pounds and afford a balanced distribution of the weight.

c. A configuration that directs the attaching cables toward the ground, to minimize strain, with sufficient clearance to ensure the NPFA 70, recommended bends for the cables are not encroached.

3.3.10 Illuminated indicators. All illuminated indicators shall meet the following secure lighting (blue/green) requirements:

a. Light sources shall consist of light emitting diodes (LEDs).

b. 95 percent of the light energy emitted by each light source shall be at wavelengths below 700 nanometers.

c. All light sources shall be visible in bright sunlight.

d. All light sources shall be compatible with NVIS Soldier equipment IAW MIL-L-85762.

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3.3.11 Illumination system. The PDISE Type V, Illumination System, single phase, shall consist of utility lights, extension cords, a branch circuit assembly, outlet boxes, etc. as described in the specification sheet (MIL-DTL-53126/5).

3.3.12 Universal adapter. The PDISE Type VI, Universal Adapter, shall enable the PDISE to interface with and provide power to any load.

3.3.13 Wiring. Wiring shall conform to NFPA 70. Where appropriate, all wires shall be secured neatly into harnesses. Wires in all harnesses shall be of the proper length and shall be so run and secured (with insulated clamps) as to protect insulation against contact with sharp corners and edges, pinching, sharp bending and twisting, abrasion due to vibration and exposure to fuel or lubricating oil. Wires shall not be spliced. All wires shall terminate in connectors, terminal lugs or tinned ends. Solder connection shall not be used except where connections are made to potentiometers, resistors, semiconductors, capacitors, and press-to-test indicator lights. A means shall be provided to prevent liquids from coming in contact with any electrical connection for all operating conditions specified herein. Not more than two terminal lugs shall be attached with any one screw on screw type terminal boards, and not more than four terminal lugs shall be attached to any one stud on stud-type terminal boards. Terminals on electrical components shall not have more than one wire attached.

3.3.14 Wire and wire marking. Wire shall be multi-stranded copper conductor, flame retardant, moisture, heat and oil resistant IAW the NFPA-70. All wire shall be numbered for easy identification. Each wire shall be identified by circuit number and terminations. The marking shall be permanently applied to the wire to withstand all specified operating conditions herein throughout the life of the set. Marking shall be within three inches of each end and at intervals of 5 inches throughout the wire length. The wire numbering scheme shall be consistent with that currently being employed by Industry Standards. All wires shall be marked using black characters. Except when used in a cable, the use of colored wire coating as a marking scheme is prohibited.

3.3.15 Treatment and painting. Treatment and painting shall be IAW MIL-DTL-53072. Specific treatment and painting requirements are as specified in the specification sheets. All interior and exterior surfaces shall have a finish coat of Chemical Agent Resistive Coating (CARC) paint IAW MIL-DTL-64159 or MIL-DTL-53039 except as specified herein or on applicable drawings. The color shall be green 383, color chip 34094IAW FED-STD-595. The internal components may be finished with the manufacturer's standard commercial color and paint, plating, or treatment. Non-permissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.3.15.1 Unpainted surfaces. The extension cables, utility lights, loose connector, non-metal containers, branch circuits, ropes and power cable assemblies shall not be painted.

3.3.16 Connectors. Input and output connectors shall be as specified in SAE-AS50151 and MIL-DTL-22992 (see TABLE II and Table III) IAW the specification sheets (MIL-DTL-53126/1, /2, /3, /4, /5) and drawings(see TABLE I). Each shall be identified by circuit number and amperage, consistent with corresponding identification on the control panel. Receptacles shall be keyed to accept only a plug of the same rating. Connectors shall be provided with tethered environmental resistant covers.

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3.3.16.1 Bulkhead connectors. Bulkhead connectors shall mate with MS90557 connectors, for input; and MS3406 and MS90556 connectors, for output as specified in the specification sheets (MIL-DTL-53126/1, /2, /3, /4). Location of connectors shall have sufficient clearance to facilitate coupling and de-coupling by personnel wearing arctic mittens, taking into account the captive caps and plugs. The bulkhead connectors shall be straight connectors, with covers, matching the following shell sizes and configurations:

TABLE II. Bulkhead connectors.

Rating	Input (plugs) shall mate with	Output (sockets) shall mate with
200 A/P	M90557C52413S	—
100 A/P	M90557C44413S	M90556C44413P
60/40 A/P	M90557C32413S	M90556C32413P
60 ampere	M90557C32405S	—
20 ampere	—	M3406D16-10S

3.3.16.2 Cable connectors. Cable connectors shall mate with MS3401, MS3402, MS90555, and MS90557 connectors, for input; and MS3406, MS90556, and MS90558 connectors, for output as specified in the specification sheets (MIL-DTL-53126/1, /2, /3, /4, /5). Cable connectors shall be straight connectors, with covers, matching the following shell sizes and configurations:

TABLE III. Cable connectors.

Rating	Input (plugs) shall mate with		Output (sockets) shall mate with	
200 A/P	—	M90557C52413S	M90558C52413P	M90556C52413P
100 A/P	M90555C44413S	M90557C44413S	M90558C44413P	M90556C44413P
60/40 A/P	M90555C32413S	M90557C32413S	M90558C32413P	M90556C32413P
60 ampere	—	M90557C32405S	M90558C32405P	M90556C32405P
20 ampere	M3402D16-10S	M3401D16-10S	—	M3406D16-10S

3.3.17 Receptacle assembly. The receptacle assembly shall consist of an input connector (see 3.3.16.2) fitted, 120V, 20 ampere, electrical lead, 12.00 ± 1.00 inches long, terminating in two dual duplex, straight blade, isolated ground, output receptacles IAW NEMA WD 6, 5-20R, housed in a weatherproof box. The receptacles shall conform to UL 943 and be wired IAW NFPA 70.

3.3.18 Cables. Cable design shall be IAW the drawings specified in the individual specification sheets. All cables shall comply with the requirements of NFPA 70, for flexible

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cords and cables and shall be IAW individual specification sheets (MIL-DTL-53126/1, /2, /3, /4, /5).

3.3.18.1 Input interface cable assembly. The input interface cable assembly shall be 4 ± 0.25 feet long, exclusive of the connector, and rated as specified herein and in the individual specification sheets (MIL-DTL-53126/1, /2, /3, /4, /5). One end of the cable assembly shall have an output connector (see 3.3.16.2) and the other end shall have the shielding stripped 14.50 ± 0.50 inches, with the individual conductors stripped 1.00 ± 0.25 inch, and protected against fraying.

3.3.18.2 200 A/P cable assembly. The 200 A/P cable assembly shall be 25 ± 0.25 feet long, exclusive of the connectors and rated as specified herein and in the individual specification sheet (MIL-DTL-53126/1). One end shall have an input connector and the opposite end an output connector (see 3.3.16.2) rated for 200 A/P service.

3.3.18.3 100 A/P cable assembly. The 100 A/P cable assembly shall be 50 ± 0.25 feet long, exclusive of the connectors and rated as specified herein and individual specification sheet (MIL-DTL-53126/2). One end shall have an input connector and the opposite end an output connector (see 3.3.16.2) rated for 100 A/P service.

3.3.18.4 40/60 A/P cable assembly. The 40/60 A/P cable assembly shall be 50 ± 0.25 feet long, exclusive of the connectors and rated as specified herein and in the individual specification sheet (MIL-DTL-53126/3). One end shall have an input connector and the opposite end an output connector (see 3.3.16.2) rated for 40/60 A/P service.

3.3.18.5 60 ampere cable assembly. The 60 ampere cable assembly shall be 100 ± 0.25 feet long, exclusive of the connectors and rated as specified herein and in the individual specification sheet (MIL-DTL-53126/4). One end shall have an input connector and the opposite end an output connector (see 3.3.16.2) rated for 60 ampere service.

3.3.18.6 20 ampere cable assembly. The 20 ampere cable assembly shall be provided in the following lengths: 50 ± 0.25 feet long and 25 ± 0.25 feet long, exclusive of the connectors and rated as specified herein and in the individual specification sheets (MIL-DTL-53126/3, /4). One end shall have an input connector and the opposite end an output connector (see 3.3.16.2) rated for 20 ampere service; both connectors shall be supplied with backshells.

3.3.18.7 Extension cable assembly. The extension cable assembly shall be 15 ± 0.25 feet long, exclusive of the connectors and rated as specified herein and in the individual specification sheets (MIL-DTL-53126/5). One end shall have an input connector and the opposite end an output connector (see 3.3.16.2) rated for 20 ampere service; both connectors shall be supplied with backshells.

3.3.18.8 Transition cable assembly. The transition cable assembly shall be a single female molded-plug terminated cable, 3 ± 0.25 feet long, IAW MIL-C-52286; rated for 120V, 20 ampere service. It shall be fitted with an input connector (see 3.3.16.2) with a backshell.

3.3.18.9 Branch circuit assembly, 12-outlet. The branch circuit assembly shall be a 120V, 20 ampere trunk, 24 ± 0.25 feet long, with three drop lead taps, 12.00 ± 1.00 inches long, spaced 96.00 ± 2.00 inches apart in the middle of the trunk, exclusive of the connectors and rated as specified herein and in the individual specification sheet (MIL-DTL-53126/5). These junctions shall be constructed with reinforced eyelet holes of sufficient strength to suspend the branch circuit assembly from installed “S” hooks. The trunk run shall have an input connector

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on one end, an output on the other, and each branch fitted with an output connector (see 3.3.16.2) with backshells. Each branch shall have a receptacle assembly IAW 3.3.17 attached to provide the branch circuit assembly.

3.3.18.10 In-line GFCI cable assembly. The in-line GFCI cable assembly shall be a maximum of 3.0 feet, inclusive of the connectors and rated as specified herein and in the individual specification sheets (MIL-DTL-53126/3, /4, /5). One end shall have an input connector and the opposite end an output connector (see 3.3.16.2) rated for 20 ampere service; both connectors shall be supplied with backshells.

3.3.19 Straps.

3.3.19.1 Cinching strap. The cinching strap shall be IAW the individual specification sheets (MIL-DTL-53126/3, /4, /5) and shall be made of hook and pile tape or other functionally equivalent materials. It shall be constructed of a length of the hook portion of the tape with a 1-inch (internal) D-ring attached at one end and a 2.00 inch length of the pile portion attached to the other. Other materials used shall permit fastening at any point along the length of the strap looped back onto itself through the D-ring, which shall be capable of being accomplished by a person (see 3.5.9) wearing arctic mittens. The overall length of the strap assembly shall be 18.00 ± 0.25 inches, and the strap shall be made from 1.00-inch wide stock. Unless otherwise specified (see 6.2), the color shall approximate camouflage green, color no. 34094 of FED-STD-595.

3.3.19.2 Single-strap carrier. The single strap carrier shall be IAW the individual specification sheets (MIL-DTL-53126/3) and shall encompass and hold cable assemblies and provide means of easy transporting of the cable assembly. It shall be made of textile webbing or other pliable material of similar strength, capable of being cinched by a self-clasping buckle attached at one end. If the material used is susceptible to fraying, means shall be employed to ensure that it will not interfere with issue through or the functioning of the buckle. The overall length of the strap portion shall be 20.75 ± 0.25 inches, and the strap shall be made from 1.50-inch wide stock. Unless otherwise specified (see 6.2), the color shall approximate camouflage green, color no. 34094 of FED-STD-595. A carrying handle, of the same material, shall be permanently attached to the strap adjacent to the buckle. The handle shall be suitable for use by personnel (see 3.5.9) wearing arctic mittens. An identification tag, of the same or similar material in white, shall be permanently attached to the strap in a conspicuous location.

3.3.19.3 Double-strap carrier. The double-strap carrier shall be IAW the individual specification sheets (MIL-DTL-53126/1, /2, /4)and shall be of the same construction as the single-strap (see 3.3.19.2) but shall consist of two straps, 27.75 ± 0.25 inches long, joined below the carrying handle to form an assembly 10.00 ± 0.50 inches across. The handle of the carrier shall be attached, one strap to the other, across the assembly.

3.3.20 Transit and storage containers. Cabling and accessories shall be stored in transit containers as specified in the individual specification sheets (MIL-DTL-53126/3, /4, /5). The containers shall be reusable and protect the cabling and accessories during all transportation and storage modes. The transit and storage container shall be sized appropriately to contain all cables and accessories as specified in the individual specification sheets (MIL-DTL-53126/3, /4, /5)

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3.4 Interface.

3.4.1 Power source compatibility. The feeder and distribution systems and electrical ancillary items shall physically inter-connect with DoD Standard Family of Generator sets, which operate at 120 volts, single phase and 120/208 volts, three phase, at 50 Hz or 60 Hz. Each system shall be usable with 120/130 volt, 50 Hz or 60 Hz, commercial utility power.Interconnection. The feeder, distribution and illumination systems shall interconnect with one another in order to form an electrical grid.

3.4.3 Universal adapter. The universal adapter shall provide a universal interface for loads that cannot interface using output connectors as specified herein.

3.5 Environmental parameters. The PDISE feeder and distribution systems shall withstand the following environmental conditions:

3.5.1 Operating temperatures. The PDISE shall perform continuously for not less than 12 hours, in any ambient temperature from -25 °F to 120 °F, without performance degradation, or mechanical or electrical failure.

3.5.2 Storage temperatures. The PDISE shall withstand indefinite storage in any ambient temperature from -50 °F to 160 °F, without subsequent performance degradation, or mechanical or electrical failure.

3.5.3 High/low pressures. The PDISE shall be operable, transportable, and storable at elevations up to 10,000 feet above sea level, and shall withstand normal atmospheric pressure changes during operation. It shall be air transportable through varying atmospheric pressures, and shall survive rapid decompression.

3.5.4 Humidity. The PDISE shall withstand prolonged storage and operation in all possible relative humidities, with ambient temperatures ranging from -50°F to 160°F (-46°C to 52°C) at sea level.

3.5.5 Fungus and moisture resistance. The PDISE shall withstand the effects of fungus and moisture. The electrical circuitry, components, and connections that are not inherently fungus and moisture resistant or hermetically sealed shall be protected from fungus growth and moisture by an appropriate coating treatment.

3.5.6 Rain. The PDISE shall operate with 5 inches (127 millimeters) of rain per hour impinging on the PDISE at angles from vertical up to 45 degrees.

3.5.7 Sand and dust. The PDISE shall operate in a sand and dust environment with particle concentrations up to 1,400 milligrams per cubic meter (mg/m^3) and particle sizes ranging from less than 74 micrometers in diameter to 1,000 micrometers with the bulk of the particles ranging in size from 74 to 350 micrometers.

3.5.8 Chemical, Biological, Radiological and Nuclear (CBRN) Contamination Survivability. The PDISE shall have a Chemical Agent Resistant Coating (CARC) painted as specified herein. The PDISE shall be capable of being serviced and maintained by personnel wearing Mission Oriented Protective Posture (MOPP) IV chemical, biological, and radiological clothing (see 6.5.4) without special tools or support equipment.

3.5.9 Human Factors Engineering (HFE). The PDISE shall be designed IAW accepted criteria of design for HFE as described in MIL-STD-1472. The PDISE shall be operable and

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maintainable during day and night by 5th percentile female through 95th percentile male soldiers when wearing the clothing, which is appropriate to the environments detailed herein.

3.5.10 Safety. The PDISE shall meet the provisions of MIL-STD-882; Hazards and safety requirements of MIL-STD-1472; and the safety design criteria specified herein. Exposed parts of such nature as to be a hazard to personnel shall be insulated, enclosed or guarded without impairing the functioning of these parts. Except where functionally required, exposed surfaces shall be free from burrs, sharp edges and corners, or other features that present a personnel safety hazard (see 6.5.1); or shall be shielded against contact. Danger or caution signs, labels, and markings (see 3.6.2) shall be used to warn of potential or specific hazards. The sign(s) shall be prominently displayed on the PDISE, and visible from any angle of approach.

3.5.11 Transportability. The PDISE shall not be damaged by rough handling (see 6.5.5) which could be encountered during rail (see 6.5.6), truck or trailer (see 6.5.7), and aircraft transportation (see 6.5.1). The utility kit and feeder/distribution systems in operational mode but without cables attached shall be soldier-portable, indefinitely, over battlefield terrain, without damage (see 6.5.3) or malfunction. The utility storage box shall be capable of being stacked 8 boxes high without damage.

3.6 Marking. The PDISE shall have marking or permanently attached data plates, in a location where the data will be prominent and legible but not interfere with PDISE operation or maintenance. Legibility shall be IAW MIL-STD-1472 (see 3.5.9). Safety and instructional markings shall not be obscured by components, to include occasionally positioned levers, etc., from the operator's position. Marking shall be of a contrasting color to the surface on which it is inscribed, and indelible for the anticipated life of the PDISE. Plates shall be constructed of a durable material that is corrosion, ozone, and weather resistant.

3.6.1 Identification. The PDISE shall have an identification plate IAW drawing 13230E7022.

3.6.2 Safety, instructional, and component labeling. Warning and caution markings shall be IAW NEMA Z 535.2. For the systems, at a minimum, a “POWER IN” decal shall be located adjacent to the indicator(s); a warning decal stating “WARNING LETHAL VOLTAGE. DISCONNECT POWER BEFORE SERVICING” shall be prominently positioned at access to the interior. The input and output connector ratings; the bus bars, connectors, and grounding points shall also be identified. The contents of the utility set shall be identified on the exterior of its storage container.

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4 VERIFICATION

4.1 Classification of inspection. Inspections as specified herein, shall be classified as follows:

a. First article preproduction inspection (see 4.3).

b. Conformance inspection (see 4.4).

c. Inspection comparison (see 4.5).

d. Inspection of packaging (see 4.6).

4.2 Test schedule. Testing shall be performed on a complete assembled PDISE. PDISE components may be networked, and tested as a single assembly. Components and other devices, or load banks shall be used to test throughput of the PDISE. Electric power shall be supplied from a regulated, three phase, 120/208 VAC, 50 or 60 Hz source. Unless otherwise specified in the contract, testing shall be performed in the sequence noted in TABLE IV.

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TABLE IV - Test schedule.

1	2	3	4	5
First Article	Conformance Inspection			
P P T SEE 4.3	I N D I SEE 4.4	TEST	TEST METHOD (MIL-STD-705) AND TEST PARAGRAPH	BASIC REQUIREMENT PARAGRAPH
X	X	Operational	4.7.1	3.1
X	X	Circuit Interrupter (Overload Current)	4.7.1.2, TM 512.2d	3.3.6, 3.3.8
X	X	Interface	4.7.1.3	3.4
X		High Temperature	TM 710.1d at 135°F (57°C) Operate for 2 hours.	3.5.1
X		Size and Weight	4.7.4	3.1
X		Safety	4.7.1.1	3.5.10
X		Human engineering	4.7.1.1	3.5.9
X		Rain	4.7.2.5; MIL-STD-810, TM 506.5	3.5.6
X		Carry	4.7.3.1	3.5.11
X		Stacking	4.7.3.2	3.1.3of MIL-DTL-53126A/5

TABLE V. - Additional test schedule.

1	2	3
TEST	TEST METHOD (MIL-STD-705) AND TEST PARAGRAPH	BASIC REQUIREMENT PARAGRAPH
Altitude test	4.7.2.3	3.5.3
Extreme Cold Storage	3.5.2; TM 731.1c	3.5.2
Drop	4.7.3.3	3.5.11
Fungus and moisture resistance	4.7.2.4; (TM 507.5, Procedure II, MIL-STD-810)	3.5.5
Hot Storage Test	4.7.2.1	3.5.2

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TABLE V. - Additional test schedule.

1	2	3
TEST	TEST METHOD (MIL-STD-705) AND TEST PARAGRAPH	BASIC REQUIREMENT PARAGRAPH
Humidity	4.7.2.4; (TM 507.5, Procedure II, MIL-STD-810)	3.5.4
Lifting and Tie-down	4.7.3; MIL-STD-209	3.3.9b, 3.5.11
Low Temperature	4.7.2.2	3.5.1
Railroad Impact	4.7.3; MIL-STD-810, TM 526	3.5.11
Sand and dust	4.7.2.6; MIL-STD-810, TM 510.5	3.5.7
Transportability	4.7.3	3.5.11

4.3 First article preproduction inspection. The preproduction PDISE shall be examined and tested as specified herein to determine compliance with this specification and the applicable specification sheets (see 3.1). This inspection shall include the examination of 4.3.1 and the tests of 4.3.2. Non-conformance to any specified requirement, the presence of one or more defects, or the failure of any test shall constitute cause for rejection.

4.3.1 Examination of PDISE. The PDISE shall be examined for defects and conformance to the TDP. The defects include those listed in TABLE VI. Circuit breaker, connector, and internal wire markings and terminations shall be verified for all circuits. Evidence that the PDISE do not comply with the drawings shall be cause for rejection of the PDISE.

4.3.1.1 Examination of components and subassemblies. Examination of components and subassemblies shall be made prior to assembly of the PDISE. Evidence of any components or subassemblies that do not comply with the requirements of this specification shall be cause for rejection of that component or subassembly.

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TABLE VI – Examination schedule.

Exam	Defect	Exam method	Reqt para
E1	Input interface cable assembly not as specified.	Visual/Doc/SIE	3.3.18.1
E2	200 A/P cable assembly not as specified.	Visual/Doc/SIE	3.3.18.2
E3	100 A/P cable assembly not as specified.	Visual/Doc/SIE	3.3.18.3
E4	40/60 A/P cable assembly not as specified.	Visual/Doc/SIE	3.3.18.4
E5	60 ampere cable assembly not as specified.	Visual/Doc/SIE	3.3.18.5
E6	20 ampere cable assembly not as specified.	Visual/Doc/SIE	3.3.18.6
E7	Extension cable assembly not as specified.	Visual/Doc/SIE	3.3.18.7
E8	Transition cable assembly not as specified.	Visual/Doc/SIE	3.3.18.8
E9	12-outlet, branch circuit, cable assembly not as specified.	Visual/Doc/SIE	3.3.18.9
E10	Cinching strap not as specified.	Visual/SIE	3.3.19.1
E11	Single-strap carrier not as specified.	Visual/SIE	3.3.19.2
E12	Double-strap carrier not as specified.	Visual/SIE	3.3.19.3
E13	Type I feeder system not as specified.	Visual/Doc/SIE	MIL-DTL-53126A/1
E14	Type II feeder system not as specified.	Visual/Doc/SIE	MIL-DTL-53126A/2
E15	Type III distribution system not as specified.	Visual/Doc/SIE	MIL-DTL-53126A/3
E16	Type IV distribution system not as specified.	Visual/Doc/SIE	MIL-DTL-53126A/4
E17	Type V utility set not as specified.	Visual/Doc/SIE	MIL-DTL-53126A/5
E18	Utility light not as specified.	Visual/Doc/SIE	MIL-DTL-53126A/ 5
E19	Support line not as specified.	Visual/Doc/SIE	MIL-DTL-53126A/5
E20	Storage container not as specified.	Visual/Doc/SIE	MIL-DTL-53126A/5
E21	Interconnection not as specified	Visual/Doc/SIE	3.4.2
E22	Human Engineering characteristics not as specified	Visual/Doc/SIE ①	3.5.9
E23	Materials not as specified	Visual/Doc	3.3

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TABLE VI – Examination schedule.

Exam	Defect	Exam method	Reqt para
E24	Marking not as specified	Visual/Doc	3.6

4.3.2 First article preproduction tests. Two preproduction PDISE of each type shall be subjected to the applicable tests marked "X" under column 1 of TABLE IV. In addition to any test specified as part of the preproduction tests, the Government reserves that right to conduct any and all preproduction tests and failures of such additional tests shall have the same effect as failures of those tests specified as preproduction tests.

4.4 Conformance inspection. Each individual production PDISE shall be examined and tested as specified herein to determine compliance with this specification and the applicable specification sheets (see 3.1). This inspection shall include the examination of 4.4.1 and the tests of 4.4.2. Non-conformance to any specified requirement, the presence of one or more defects, or the failure of any test shall constitute cause for rejection.

4.4.1 Conformance examination of PDISE. The PDISE shall be examined for defects and conformance to the TDP. The defects include those listed in TABLE VI. Circuit breaker, connector, and internal wire markings and terminations shall be verified for all circuits. Evidence that the PDISE do not comply with the drawings shall be cause for rejection of the PDISE.

4.4.1.1 Conformance examination of components and subassemblies. Examination of components and subassemblies shall be made prior to assembly of the PDISE. Evidence that any components or subassemblies do not comply with the requirements of this specification shall be cause for rejection of that component or subassembly.

4.4.2 Conformance tests. Each individual production PDISE shall be subjected to the applicable tests marked "X" under column 2 of TABLE IV. In addition to any test specified as part of the conformance tests, the Government reserves that right to conduct any and all conformance tests and failures of such additional tests shall have the same effect as failures of those tests specified as conformance tests

4.5 Inspection comparison. The Government may select PDISE at any time during the contract production period and subject these PDISE to any examination and test specified herein, necessary to determine that the selected PDISE meet all requirements of this specification. The inspection shall be performed by the Government at a site selected by the Government. PDISE will be selected at random from those that have been accepted by the Government, and shall not include the previously inspected preproduction and production PDISE. Acceptance of an inspection comparison PDISE shall not exclude the remaining PDISE from the quality conformance inspection and acceptance provisions specified in section 4.4. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.5.1 Inspection failure. Failure of an inspection comparison PDISE to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.5,

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shall be cause for refusal by the Government to continue acceptance of production PDISE until the contractor has provided evidence that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on PDISE previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all PDISE accepted prior to completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting office.

4.6 Inspection of packaging,

4.7 Tests. Testing shall be performed on a complete, assembled PDISE that shall include, at a minimum, one each of every component of the utility kit connected to the power distribution system under test. Feeder systems, distribution systems, and utility set components may be networked, and tested as a single assembly. Singular feeder system testing shall utilize a distribution system, or another method devised, to accomplish connection of the utility set components. Components and other devices, or load banks shall be used to test throughput of the feeder/distribution systems. Electric power shall be supplied from a regulated, three phase, 120/130 VAC, 50 or 60 Hz source. Testing shall be conducted without cover, at any ambient conditions permitted in 3.5, except as required for environmental tests.

4.7.1 Operational test. During this procedure, every input circuit breaker shall be operated at its rated load for not less than 4 continuous hours, and every output circuit breaker shall be operated at its rated load for not less than 2 continuous hours. Every cable, receptacle assembly, and other electrical component of the end item shall be utilized or operated at its rated capacity for not less than 1 continuous hour. All switches shall be turned on and off under rated load not less than twice to ascertain functionality. Inability to provide a sustained transfer of capacities specified or other nonconformance to the requirements specified herein and applicable specification sheets, shall constitute failure of this test. Any lack of ground continuity or instance of open or short circuits, safety hazard (see 6.5.2), damage (see 6.5.3) to components, or malfunctions that reduce the specified efficiency of the system or any component shall also constitute failure of this test.

4.7.1.1 Human engineering and safety examination. During the operational test(s), the design shall be evaluated for conformance to the human factors, accessibility, and safety requirements specified in 3.5.9 (see C.A.1).

4.7.1.2 Circuit Interrupter (Overload Current). During the operational test(s), the circuit interrupter shall be tested to determine compliance with specified overload protection. The test shall be conducted IAW TM 512.2d, MIL-STD-705 for each circuit breaker in the feeder or distribution system.

4.7.1.3 Interface. Proper interface shall be determined during operational test. Each item shall be interconnected to determine proper form, fit and function.

4.7.1.4 Continuity test. The continuity of each circuit shall be tested with an ohmmeter. A resistance greater than 1 ohm shall constitute failure of this test.

4.7.1.5 Bonding test. The continuity of the housing, ground bus bar, and connectors to ground shall be tested with an ohmmeter. A resistance greater than 1 ohm shall constitute failure of this test.

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4.7.2 Environmental. During the subordinate temperature tests, the Type V PDISE utility lights shall be connected to the PDISE feeder/distribution system under test, and operated for not less than 30 minutes of each test. The M46 storage box shall be opened and shall remain open, unassisted, for not less than 15 minutes of each test.

4.7.2.1 High temperature. The PDISE shall be tested IAW MIL-STD-810, TM 501.5, Procedure I and II. The PDISE shall be soaked at a temperature of 160 °F, without solar radiation, for not less than 24 hours; then the temperature shall be lowered to 120 °F. After unit temperature stabilization at 120 °F, the Type I through IV and V PDISE shall be connected to the power source and operated at rated capacity for not less than 12 hours. In addition to failure criteria specified in MIL-STD-810, nonconformance to 3.5.1 and 3.5.2 shall constitute failure of this test. Failure of the lights to operate, or the box to remain open shall also constitute failure of this test, for the Type V PDISE only.

4.7.2.2 Low temperature. The PDISE shall be tested IAW MIL-STD-810, TM 502.5, Procedure I, II, and III. The PDISE shall be soaked at a temperature of -28 °F, without solar radiation, for not less than 24 hours; then the temperature shall be raised to -25 °F. After unit temperature stabilization at -25 °F, the Type I through IV and VI PDISE shall be connected to the power source and operated at rated capacity for not less than 12 hours. In addition to failure criteria specified in MIL-STD-810, nonconformance to 3.5.1 and 3.5.2 shall constitute failure of this test. Failure of the lights to operate, or the box to remain open shall also constitute failure of this test, for the Type V PDISE only.

4.7.2.3 High/low pressure. The PDISE shall be tested IAW MIL-STD-810, TM 500.5, Procedure I, II, and III in an atmospheric pressure (altitude simulating) chamber selected by the contractor and approved by the Government. The altitude change rate shall not be greater than 2,000 ft/min, with the exception of the rapid/explosive decompression tests, and the PDISE tested shall be tested at between 65 °F and 95 °F:

a. Procedure I – The test shall be conducted at a simulated altitude of 50,000 ft, with the PDISE in its transport configuration. Atmospheric pressure shall be stabilized and maintained for not less than 1 hour. The PDISE shall then be returned to local ambient pressure and the Type I through IV PDISE tested as specified in 4.7.1.

b. Procedure II – The test shall be conducted at a simulated altitude of 15,000 ft. Atmospheric pressure shall be stabilized and maintained for 6 hours with the Type I through IV PDISE operational during the first and last hours of the test.

c. Procedure III – The test shall be conducted from a simulated altitude of 50,000 ft, with the PDISE in its transport configuration. Atmospheric pressure shall be stabilized and the PDISE subjected to a near-instantaneous decompression rate, to ambient, of 0.1 second.

In addition to failure criteria specified in MIL-STD-810, failure to operate as prescribed herein, nonconformance to 3.5.3, or the PDISE presenting a hazard shall constitute failure of this test.

4.7.2.4 Humidity. The PDISE shall be tested IAW MIL-STD-810, TM 507.5, Procedure II, Category (a) and (c). The PDISE shall be placed in a test facility selected by the contractor and approved by the Government. Using a standard aggravated temperature-humidity 24-hour cycle, the duration of the humidity test shall be a total of 16 days, with the PDISE in its normal operating configuration. The quick look schedule shall entail an operational check, of

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not less than 30 minutes, after the first 24-hour period and at least once every 5 days thereafter, at regular intervals. At the conclusion of the exposure, the PDISE shall be stabilized at ambient conditions and tested as specified in 4.7.1. In addition to failure criteria specified in MIL-STD-810, failure to operate as prescribed herein, nonconformance to 3.5.3, or the sustainment of fungus growth shall constitute failure of this test.

4.7.2.5 Rain. The PDISE shall be tested IAW MIL-STD-810, TM 506.5, Procedure I. The test shall consist of simulating rain and wind-blown rain onto the PDISE in its operational and storage configurations. Nozzles used shall produce a square spray pattern or other overlapping pattern, and a droplet size predominantly in the 0.08 to 0.18 inch range at 40 ± 5 pounds per square inch, gage (psig). At least one nozzle shall be used for each 6 foot square area and positioned 19.00 ± 1.00 inches from the test surface. The wind source shall be located, with respect to the test item, so that is will cause rain to beat directly and uniformly against one side of the PDISE, with variations up to 45° from the horizontal. The wind source shall be capable of producing wind velocities to 65 mph. Each PDISE shall be exposed to the rain simulation for not less than 40 minutes per face, at 10-minute intervals of 15, 25, 45, and 65 mph wind velocities. At the end of the inspection, the Type I through IV PDISE shall be connected to the power source, and operated at rated capacity for not less than for 30 minutes. Any penetration of water into the case (Type V) or any accumulation of water greater than 4 cm^3/28000 cm^3 (Type I through IV) shall constitute failure of this test. In addition to other failure criteria specified in MIL-STD-810, failure of the feeder/distribution PDISE to operate as specified herein shall also constitute failure of this test.

4.7.2.6 Sand and dust. The PDISE shall be tested IAW MIL-STD-810, TM 510.5, to verify its withstanding the effects of sand and dust. Testing shall be conducted in an environmental chamber capable of controlling particulate contamination and distribution. The order of these subordinate tests is optional, but they shall be conducted on the same PDISE(s) and within 24 hours of each other. In addition to failure criteria specified in MIL-STD-810, failure of the Type I through IV PDISE to operate as specified herein or abnormally high operating temperatures shall constitute failure of this test.

4.7.2.6.1 Sand. The composition used for testing shall be silica sand and its concentration that for equipment operated close to aircraft on unpaved surfaces, as specified in MIL-STD-810. The air velocity shall be greater than 3540 ft/min and the PDISE shall be exposed for a total of not less than 360 minutes, at 90 ± 5 minutes per side. At the end of the exposure, the PDISE shall be inspected for damage, and the Type I through IV PDISE connected to the power source, and operated at rated capacity for not less than a period of 30 minutes.

4.7.2.6.2 Dust. The composition used for testing shall be red china clay or silica flour and its concentration that for equipment operated close to aircraft on unpaved surfaces, as specified in MIL-STD-810. The air velocity shall be greater than 1750 ft/min and the PDISE exposed for not less than six hours. At the end of the exposure, the PDISE shall be inspected for damage, and the Type I through IV PDISE connected to the power source, and operated at rated capacity for not less than 30 minutes.

4.7.3 Transportability. The PDISE shall be mounted onto a (railcar) flatbed and restrained as recommended in MIL-STD-209. The flatbed shall be propelled to a speed of not less than 25 miles per hour and, at that speed, be permitted to impact an immovable object.

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Nonconformance to 3.5.11, and as tested in 4.7.1 for the distribution/feeder systems, shall constitute failure of this test.

4.7.3.1 Carry. The Type V PDISE shall be weighted to not less than 200% of its normal operational weight and suspended by its handles. It shall be raised to a height of not less than three feet from the base of the box to the ground and be allowed to free fall for 2 feet and then remain suspended, only by its handles, for a period of not less than 30 minutes. Any separation or damage to components, detrimental deformation, or nonconformance to 3.5.11, and as tested in 4.7.1 for the distribution/feeder systems, shall constitute failure of this test.

4.7.3.2 Stacking. Eight Type V PDISE shall be weighted as necessary to achieve an individual gross weight of 125 pounds and stacked one atop the other. They shall be subjected to an environment of 120 °F, at a relative humidity of not less than 60 percent, for a period of not less than 24 hours. At the end of this period, the second PDISE from the bottom of the stack shall be removed and inspected. Any separation or damage to components, detrimental deformation, or nonconformance to 3.1.3 of MIL-DTL-53126A/5 shall constitute failure of this test.

4.7.3.3 Drop test. The PDISE shall be drop tested IAW MIL-STD-810, TM 516.6, Procedure IV to verify the structural integrity and functional integrity of the PDISE subjected to a transit drop. The drop heights for each individual PDISE shall be determined by the weight specified in the individual specification sheets (MIL-DTL-53126/1, /2, /3, /4, /5) and the drop height specified in MIL-STD-810, Table 516.6-IV. At the end of the drop, the PDISE shall be inspected for damage, and the Type I through IV PDISE and Type VI connected to the power source, and operated at rated capacity for not less than 30 minutes.

4.7.4 Size and weight. The overall size shall be determined with all of the PDISE doors in the closed position, using standard measuring equipment. The PDISE shall be weighed to determine the weight, using a calibrated scale.

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5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6 NOTES

(This section contains information of a general or explanatory nature, which may be helpful but is not mandatory.)

6.1 Intended use. The PDISE equipment will be employed to meet the power distribution and illumination requirements of combat support and combat service units. The feeder systems will be used to transmit electricity through a series of cables, distribution centers, and branch circuits from generator sets, or commercial power sources, to electrically operated equipment located about and within the tents of the units. The system is military unique in that there are no commercial equivalents to the systems. Further, individual components that may be procured and assembled to produce the items will not meet the interface, environmental, or operational requirements of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, required (see 1.2.1).

c. When a first article is required for inspection and approval, time frame for submission and the number of units required (see 3.1.1 and 6.3).

d. When color will be other than as specified (see 3.3.19**,** 3.1.1, and 3.3.15.1).

e. Packaging requirements as listed in Appendix B

6.3 First article. When a first article inspection is required, the PDISE should be a preproduction model. The first article should consist of one or more of each end item and supporting components for distribution equipment specified. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article test results and disposition of the documents' first article. Invitations for bid should provide that the Government reserves the right to waive the requirement for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, illustrated parts lists, maintenance and operation manuals and other instructional materials to be furnished with each PDISE.

6.5 Definition. The following definitions apply throughout this specification.

6.5.1 Aircraft and helicopter transportation. Aircraft and helicopter transportation are to be interpreted to mean a 9 inch (22.86 cm) end drop under test conditions specified in MIL-STD-705, TM 740.3.

6.5.2 Safety hazard. Safety hazards may be interpreted as defined in MIL-HDBK-454, Guideline 1, Safety Design Criteria-Personnel Hazards.

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6.5.3 Damage. Damage is discoloring, swelling, corrosion, melting, permanent setting or loss of resiliency in elastomers, binding of parts, deformation, loosening, breakage, cracking, change of fit of any component or part, or any condition resulting in malfunction of the PDISE or any component thereof.

6.5.4 Mission orientated protective posture (MOPP). MOPP I protection consists of a two-piece protective over garment and helmet (in hot weather the jacket can be open with trousers closed). MOPP IV protection consists of a two-piece protective over garment, hood, over-boots and rubber gloves with cotton liners. The over garment is closed and hood is pulled down and adjusted IAW Field Manual FM 3-11.4.

6.5.5 Rough handling damage. Rough handling damage is defined as any condition resulting in malfunctioning of the unit, liquid leakage, deformation, loosening, breakage, or change of fit of any component or part.

6.5.6 Railroad transportation. Railroad transportation are to be interpreted to mean successfully completing the railroad impact test and post-test functional checks (see 4.7.3).

6.5.7 Truck or trailer transportation. Truck or trailer transportation is defined as the conditions encountered during the following road tests (see 4.7.3):

a. Four cycles of the off-road, cross-country transport test, each cycle consisting of the following road conditions (see TABLE VII):

TABLE VII – Road condition test specifications

Road Condition	Distance mi (km)	Speed mph (km/h)
Paved Highway (Perryman Paved)	250 (402.3)	up to 50 (80.5)
Level Cross Country (Perryman 1)	250 (402.3)	up to 20 (32.2)
Hilly Cross Country (Churchville B)	125 (201.1)	up to 20 (32.2)
Belgian Block (Munson Test Area)	15 (24.1)	up to 20 (32.2)

b. Five cycles of the Munson Test Area (MTA) test courses, each cycle consisting of the following test course (see TABLE VIII):

TABLE VIII – MTA test course specifications

MTA Test Course	Speed in mph (km/h)
Belgian Block	20 (32.2)
2-Inch Washboard	10 (16.1)
3-Inch Spaced Bump	20 (32.2)
Radial Washboard	15 (24.1)
6-Inch Washboard	5 (8.0)

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6.6 Subject term (key word) listing.

Circuit
Feeder System
Conductor
Lighting

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes

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APPENDIX A

TEST METHODS

GENERAL REQUIREMENTS

A.1 SCOPE

A.1.1 Scope. This appendix is a mandatory part of the specification. The information contained herein is intended for guidance only. This document clarifies the test methods (TMs) to be used for testing the PDISE.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

STANDARDS

MILITARY

MIL-STD-882	System Safety Program Requirements.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems.

HANDBOOKS

MILITARY

MIL-HDBK-705	Generator Sets, Electrical, Measurement and Instrumentation Methods

A.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ISO 10012	Quality Assurance For Measuring Equipment, Part I, Meteorological Confirmation System for Measuring Equipment.
NCSL Z 540-1	General Requirements for Calibration Laboratories, Measuring and Test Equipment.

(Application for copies should be addressed to: American National Standards Institute, 1430 Broadway, New York, NY 10018 or copies are available at: www.ansi.org.)

A.3 GENERAL REQUIREMENTS

A.3.1 Test conditions. Unless otherwise specified herein, measurements and tests shall be made at the following conditions:

a. Standard ambient. Ambient measurements and checks (e.g., pre- and post-test) are conducted at room ambient conditions as follows:

Temperature: 77±18ºF (25±10ºC)

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Relative humidity:	Uncontrolled room ambient
Atmospheric pressure:	Site pressure

b. Controlled ambient. When the ambient conditions must be closely controlled, the following shall be maintained:

Temperature:	73±3.6ºF (23±2ºC)
Relative humidity:	50±5%
Atmospheric pressure:	86.45 to 103.05 kPa
	(655 to 775 mmHg)
	(25.5 to 30.5 inHg)

A.3.1.1 Tolerance for test conditions. Unless otherwise specified, tolerances for test conditions shall be as follows:

a. Temperature. The test item shall be totally surrounded by an envelope of air (except at necessary support points). The temperature of the test section measurement system and the temperature gradient throughout this envelope, which is measured close to the test item, shall be within ±3.6ºF (±2ºC) of the test temperature and shall not exceed 1.8°F (1ºC) per meter or a maximum of 4.0°F (2.2ºC) total (equipment not operating).

b. Pressure. ±5% (±200 Pa).

c. Humidity. Relative humidity at the chamber control sensor shall be ±5% RH of the measured value.

d. Time. Elapsed time shall be measured with an accuracy of ±1%.

e. Air velocity. Air velocity shall be within 10% of the specified value.

A.3.1.2 Accuracy of test instrumentation calibration. The accuracy of instruments and test equipment used to control or monitor the test parameters shall be verified prior to and following each test and then calibrated in predetermined intervals and shall meet the requirements of ISO 10012 and ISO/IEC 17025 to the satisfaction of the procuring activity. All instruments and test equipment used in conducting the tests specified herein shall:

a. Be calibrated to laboratory standards whose calibration is traceable to the National Standards via primary standards.

b. Have an accuracy of at least one third the tolerance for the variable to be measured. In the event of a conflict between this accuracy and a requirement for accuracy in any one of the TMs of this appendix, the latter shall govern.

A.3.2 General test performance guidance.

A.3.2.1 Pretest performance record. Before testing, the test item should be operated at standard ambient conditions to obtain and record data determining compliance with the requirements document(s) and for comparison with data obtained before, during, and after the environmental test(s). The identification and environmental test history of the specific test

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item(s) should be documented for failure analysis purposes. The pre-test record shall include (as applicable):

a. The functional parameters to be monitored during and after the test if not specified herein. This shall include acceptable functional limits (with permissible degradation) when operation of the test item is required.

b. Additional evaluation criteria.

A.3.2.2 Installation of test item in test facility. Unless otherwise specified, the test item shall be installed in the test facility in a manner that will simulate service usage, with connections made and instrumentation attached as necessary.

a. Plugs, covers, and inspection plates not used in operation, but used in servicing, shall remain in place.

b. Electrical connections normally used in service but not in test shall be provided with electrical connectors having dummy cables with protected terminations. Such mechanical connections shall also be protected.

c. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test item is installed or as specified in the individual TMs.

d. The test item shall be operated according to the applicable technical order or technical manual, when available, to determine that no malfunction or damage has resulted from faulty installation or handling. The requirement to operate the test item after its installation in the test facility applies only when the item is required to operate during the test.

e. Test items shall be positioned at least 6 inches (15 cm) from each other or from walls, floors, ceilings, etc. to allow for adequate circulation.

f. If the item to be tested consists of several separate units, these units may be tested separately provided the functional aspects are maintained as defined herein.

A.3.2.3 Interrupted tests. Unless otherwise specified in the individual TMs, the following procedures shall be followed when a test is interrupted. Any deviation from this guidance shall be explained in the test report.

A.3.2.3.1 In-tolerance interruptions. Interruptions during which the prescribed test tolerances are not exceeded shall be considered as part of the total test duration. (No allowance is necessary if exposure to the proper test levels was maintained.)

A.3.2.4 Post-test data. At the completion of each test, the test item shall be inspected as specified herein, and the results shall be compared with the pretest data obtained in accordance with A.3.2.1. Post-test data shall include (when applicable):

a. Complete identification of all test equipment and accessories.

b. The actual test sequence (program) used.

c. Deviation from the planned test program.

d. The room ambient test conditions recorded periodically during the test period.

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e. A signature and data block for certification of the test data by the test engineer.

f. Other data as specified in the individual TMs or as specified herein.

A.3.2.5 Failure criteria. Failure of the test item to meet any one of the following conditions shall constitute a test item failure.

a. Deviation of monitored functional parameter levels beyond acceptable limits established in A.3.2.1a and specified herein.

b. Non-fulfillment of safety requirements or the development of safety hazards.

c. Non-fulfillment of specific test item requirements.

d. Changes to the test item which could prevent the equipment from meeting its intended service life or maintenance requirements. (For example: Corroded oil drain plug cannot be removed with specified tools.)

Deviation from established environmental requirements.

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APPENDIX A

TEST METHODS

TEST METHOD 102.1

SAFETY AND HEALTH TEST

102.1.1 General. The generator sets and PUs must not pose an unsafe or hazardous condition to personnel.

102.1.2 Apparatus. No specific equipment is required.

102.1.3 Procedure.

102.1.3.1 Preliminary safety and health assessment. A preliminary safety and health assessment will be accomplished upon receipt of the test units. This will establish the safety of the units so that hazards to test participants can be minimized. As a minimum this preliminary assessment will include the following essentials:

a. A thorough examination will be conducted during the initial inspection and service test to identify all obvious safety problems. Based on a visual examination of the units, the Safety Checklist (TABLE 102.1- I) will be completed by the test director and, if available, a qualified safety engineer. At this point, based on the limited visual inspection, it is realized that only safety problems that are reasonably obvious will be determined since a more thorough evaluation of safety and health will be conducted during the remaining scope of testing. A cursory review of the safety and health warnings and other information will be made of the equipment manuals provided. This is to assure that sufficient information has been provided in these manuals related to safety and health, specifically in the area of data plates, information, warning plates (including noise hazard warnings if required), proper grounding procedures, lifting and tie down information, electrical shock hazard warnings, and other essential safety and health information that would ordinarily appear in a manual for a generator set. Some limited safety related tasks will be performed, if necessary, to ensure that the test items are safe for further testing.

b. During the initial inspection, an initial safety review will be made using the safety checklist contained in TABLE 102.1- I. The appropriate questions will be answered insofar as possible without further testing. Detailed comments and answers will be indicated for each applicable question of the safety checklists for electrical and mechanical hazards. Responses will not be limited to only the items appearing on the checklist; any other safety and health hazards found will be recorded and incorporated into this subtest.

102.1.3.2 Comprehensive safety and health assessment. A comprehensive safety and health assessment will be conducted throughout the entire test program. This assessment will include:

a. Systematic observations and analysis of the test units throughout all phases of the entire test to identify and investigate any actual or potential hazards to personnel and equipment that may result from operation and maintenance. A detailed description of all safety hazards identified during operation, maintenance, and all other phases of operation will be documented.

b. To ensure that the checklist of TABLE 102.1- I has been completed and that all hazards have been identified, the test units will be continuously monitored for hazards during all

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phases of testing and maintenance. It is expected that by the end of testing, all items on the safety checklist will have been answered.

c. Limitations or compromises on operating performance and maintenance because of safety considerations.

d. Examination of safety instructions and warning plates for adequacy and appropriate location.

e. Examination of operation and service manuals for adequate safety guidance concerning operation and maintenance.

f. Classification of all safety hazards in accordance with MIL-STD-882 with recommendations for appropriate corrective measures and ways to either reduce or eliminate the hazard severity and hazard probability.

102.1.4 Results.

a. List all mechanical and electrical hazards. Identify all hazards that require warning labels/placards.

b. Each NO answer reported in the checklist of TABLE 102.1- I and any other hazards identified will be reviewed and assessed to determine the degree of noncompliance with the criteria. The hazard classification (including hazard severity and hazard probability) outlined in MIL-STD-882 will be used to classify all identified safety hazards into hazard level categories. The hazards will be categorized as a deficiency, shortcoming, or suggested improvement using TABLE 102.1-II.

c. All problems recorded that have an effect on safety will be thoroughly analyzed to determine the extent of the problems and their impact on the operators, maintainers, and other personnel associated with the units. Noncompliance with specific elements of the criteria will be evaluated and suggested corrective actions will be proposed. Appropriate recommendations regarding methods to control, downgrade, or eliminate actual or potential hazards will be made so that the necessary changes can be incorporated before field deployment.

TABLE 102.1- I – Safety checklist

No	Item	Yes	No	NA	Remarks
1	Are all external parts, surfaces, shields, and all other electrically neutral parts at ground potential at all times during normal operation?				
2	Is the ground connection for all external parts mechanically secured?				
3	Is there a suitable terminal lug or other ground connection located on the chassis or frame to provide a continuous and permanent path to ground?				
4	Are grounding rods furnished?				
5	Are output terminals or other high potentials, in excess of 70 V rms, sufficiently shielded or guarded to prevent accidental contact by personnel?				
6	Are energized components located or enclosed so that suitable protection is provided against contact with un-insulated items?				
7	Are components, conductors, and shielding appropriately located such that overheating, arcing, shorting, and contact with moving parts is avoided?				
8	Are wires and cables adequately supported and terminated to prevent shock and fire hazard?				

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9	Are wires and cables properly protected against rubbing at access ports by insulated bushings?				
10	Is the test unit provided with warning placards or caution plates mounted conspicuously adjacent to any condition presenting a potential hazard to personnel (such as high voltage, rotating parts, sharp corners, etc.)?				
11	Are electrical connectors designed to ensure that only the correct plug can be inserted into its receptacle and not into a wrong receptacle?				
12	Where design considerations require plugs and receptacles of similar configuration are mating plugs and receptacle suitably coded or marked to indicate the correct mating connection?				
13	Are exposed connector pins energized after being disconnected?				
14	Are controls located away from high voltage areas?				
15	Are emergency controls placed in readily accessible positions?				
16	Is the main circuit breaker in an easily accessible location?				
17	Does a battle short switch (to bypass the safety interlocks) exist on the main control panel?				

TABLE 102.1- II – Safety checklist – Continued

No	Item	Yes	No	NA	Remarks
18	Is the battle short switch designed with a readily visible indicator light to show that it is on?				
19	Are the following protective devices present with suitable indicators to safeguard against operator injury and/or equipment failure? : Low oil pressure High coolant temperature Over-speed Low fuel Short circuit Overload Under voltage Under frequency Reverse power Over voltage				
20	Are DC power connections clearly marked for polarity?				
21	Does a DC circuit breaker exists that can cut off all power to the entire system?				
22	Are potential electrical hazards adequately treated in the instruction manual?				
23	Are operator means of detecting hazardous conditions adequate?				
24	Are circuit breakers and all control panel instruments and controls properly labeled?				
25	Does the convenience outlet have provisions for automatic grounding?				
26	Are adjustment screws or other commonly worked on parts located away from unprotected high voltages?				
27	Are tools to be used near high voltages, such as the load terminal wrench, adequately insulated?				
28	Is the grounding conductor of the equipment electrically insulated from the AC power return (neutral)?				
29	Are internal controls located at safe distances from dangerous voltages?				
30	Are protective guards sufficiently separated from exposed conductors to prevent shorting or arcing?				
31	Are components that sustain high operating temperatures during normal operation (such as exhaust pipes, turbochargers, radiators, etc.) sufficiently protected to prevent accidental contact by personnel?				

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32	Are these components adequately identified by warning plates?				
33	Are the materials used in the engine and generator housing including noise attenuating material inherently nonflammable and non-explosive?				
34	Do exposed gears, cams, levers, fans, belts, or other reciprocating, rotating, or moving mechanical parts have adequate safety covers?				
35	Are doors, hinged covers, panels, and any other exposed sharp projections or overhanging edges presenting a potential safety hazard rounded to prevent injury to personnel?				
36	Are fasteners and methods of securing doors and peripheral ancillary components sufficiently strong to prevent breakaway during normal use?				
37	Is the method of opening doors or covers evident from the construction of the cover? If not, is an instruction plate permanently attached to the outside of the cover?				
38	Is it evident when a cover is in place but not secured?				
39	Can maintenance be accomplished with shielding in place?				
40	Do external or internal surfaces that expand during maintenance have sharp edges?				
41	Is the center of gravity and weight of the test unit distinctly marked?				
42	Are weight capacities indicated on tie-downs, lifting points, etc.?				
43	Is the test unit provided with sufficient caution plates to warn maintenance personnel of potential safety hazards?				
44	Is the control panel adequately illuminated for safe and efficient operation?				
45	Have fire extinguishing methods been included in the technical publications?				

TABLE 102.1- III – Safety checklist - Continued

No	Item	Yes	No	NA	Remarks
46	Are potential mechanical hazards adequately treated in the draft instruction manual?				
47	Do floor surfaces provide adequate nonslip characteristics?				
48	Are lifting rings or slings provided?				
49	Are climbing rings, handholds, rails, etc., provided where needed?				
50	Do doors and hinged covers have positive-auction hold-open devices?				
51	Are handles recessed rather than extended where they might be hazardous?				
52	Are doors and other openings free of hazards from improperly designed catches, hinges, supports, fasteners, and stops?				
53	Are the tasks of operation and maintenance such that they do not require excessive physical strength?				
54	When glass is used is it glare-proof and shatter- proof?				
55	Does the ventilating system provide for operator safety by ducting excess heat liberated by the radiator cooling air or other hot air outlets to the exterior of the set?				
56	Are adequate precautions made to prevent exposure of operators and maintainers to exhaust gases or other toxic fumes?				
57	Is the air intake isolated or at a sufficient distance from the exhaust?				
58	Does the instruction and maintenance manual specify type of cleaning fluid and precautions to be taken when cleaning the equipment?				
59	Does the instruction and maintenance manual specify type of cleaning fluid and precautions to be taken when cleaning the equipment?				

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TABLE 102.1- IV - Hazard probability versus hazard severity

	HAZARD PROBABILITY				
	FREQUENT	**PROBABLE**	**OCCASIONAL**	**REMOTE**	**IMPROBABLE**
SPECIFIC INDIVIDUAL ITEM	Likely to occur frequently	Will occur several times in life of item	Likely to occur sometime in the life of item	Unlikely but possible to occur in the life of an item	So unlikely it can be assumed the occurrence may not be experienced
FLEET OR INVENTORY	Continuously experienced	Will occur frequently	Will occur several times	Unlikely but can reasonably be expected to occur	Unlikely to occur but possible
HAZARD SEVERITY	A	B	C	D	E
I - CATASTROPHIC May cause death or system loss	Deficiency	Deficiency	Deficiency	Deficiency	Shortcoming
II - CRITICAL May cause severe injury occupational illness, or major system damage	Deficiency	Deficiency	Deficiency	Shortcoming	Suggested improvement
III - MARGINAL May cause minor injury, minor occupational illness or minor system damage	Deficiency	Shortcoming	Shortcoming	Suggested improvement	Suggested improvement or Acceptable
IV – NEGLIGIBLE May cause less than minor injury occupational illness or system damage	Shortcoming	Suggested improvement	Acceptable	Acceptable	Acceptable

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APPENDIX A

TEST METHODS

TEST METHOD 103.1

HUMAN FACTORS ENGINEERING TEST

103.1.1 General. The generator sets and PUs must satisfy general human engineering design criteria and practices.

103.1.2 Apparatus. The following equipment shall be required: photographic and video equipment, tape measure, ruler, and illumination meter (lighted displays).

103.1.3 Procedure.

103.1.3.1 Control, display, labeling. Observations will be made of all controls, displays, and labeling with respect to HFE design practices (based on control/display integration, visual display, controls, and labeling requirements of MIL-STD-1472). Control separation and control dimensional measurements will be taken to determine if any design problems exist. The ability of the operator to successfully operate the sets while wearing regular, arctic and NBC gloves will be observed as well as the operator's ability to operate the set at night with or without wearing NVIS.

103.1.3.2 Workspace and maintenance access. Maintenance access openings and workspaces will be observed with respect to the ability of the crew to perform maintenance and to determine compatibility with anthropometrical dimensions for the 5th through 95th percentile personnel while outfitted in Army Combat Uniform (ACU), arctic, and NBC protective ensembles.

103.1.3.3 Subjective assessment.

a. Questionnaires/interviews. Human factors questionnaires will be administered to personnel assigned to the testing program. Questionnaires will primarily pertain to operating and maintaining the equipment. A section of each questionnaire will be devoted to task performance while wearing nuclear, biological, chemical (NBC) and arctic gear. Questionnaires, (TABLE 103.1- I), will be administered near the end of the test cycle to assure that all personnel are thoroughly experienced with system operations before completing the forms. Interviews will be conducted to determine the test participant's opinions on the overall operation, maintenance, and performance of the generator.

b. Checklists. Checklists (TABLE 103.1- III) will be prepared by an HFE engineer on the following elements of system design:

(1) HFE design - controls, displays, and markings.

(2) Maintainability.

c. General HFE observations. Observations will be made throughout all testing to gain additional information on any HFE-related problems. Comments and informal interviews, in addition to HFE observations, will be documented throughout to provide subjective input to assess

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the LAMPS sets and PUs. These interviews, comments, and observations will be used to augment data from other HFE subtests supplements and will be integrated into the analysis of the LAMPS sets and PUs.

103.1.3.4 Anthropometrics and demographic data.

a. Anthropometrics data. Anthropometrical measurements (standard distribution presented in TABLE 103.1- I) will be taken of test personnel who participate in daily operation, maintenance, and performance exercises.

TABLE 103.1- I– Anthropometrics and demographic data

Measurements	5th Percentile				95th Percentile			
	Regular		Arctic		Regular		Arctic	
	in	cm	in	cm	in	Cm	in	cm
Stature (nude)	64.5	163.8	-	-	73.2	186.0	-	-
Functional Reach	28.6	72.6	30.6	77.7	35.8	90.9	37.6	95.5
Sitting height, erect	33.5	85.1	34.6	87.9	38.2	97.0	`40.0	101.6
Eye height, sitting	28.6	72.6	29.4	74.7	33.3	84.6	33.8	85.9
Knee Height, sitting	19.6	49.8	22.2	56.4	23.1	58.7	25.2	64.0
Buttock-Knee length	21.6	54.9	23.6	60.00	25.3	64.3	26.5	67.3
Hip Breadth, standing	12.1	30.7	16.2	41.1	15.1	38.4	19.2	48.8
Shoulder Breadth, sitting	16.3	41.4	18.7	47.5	19.6	49.8	22.0	55.9
Buttock-popliteal length	18.1	46.0	17.2	43.7	21.5	54.6	20.5	52.0
Hand Length	6.9	17.5	8.2	20.8	8.1	20.6	9.4	23.9
Palm Length	3.1	8.0	5.0	12.7	3.8	9.7	4.8	12.2
Weight (kg)	-	57.3	-	-	-	91.6	-	-

b. Demographic data. Demographic data will be compiled of the same test participants and will include the following:

(1) Sex.

(2) Job position (for this test).

(3) Length of experience (in job position).

(4) Age.

103.1.3.5 Operator/maintainer performance tasks. The ability of the test participants to perform critical maintenance tasks while outfitted in MOPP IV and nuclear, biological, chemical (NBC) protective ensembles (arctic mittens, and NBC gloves with liners), will be determined by comparing performance times required to complete the following tasks (Excessive maintenance times or inability to perform any critical maintenance tasks will be recorded.):

a. Checking, filling, and draining engine oil.

b. Replacing engine oil filter(s).

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c. Replacing air filter element.

d. Connecting load cable.

e. Replacing fuel filter(s).

f. Checking, filling, and draining hydraulic fluid (if equipped).

g. Replacing hydraulic fluid filter(s) (if equipped).

h. Other common field maintenance actions such as filling fuel tank, radiator, battery, and adjusting belts, etc.

103.1.3.6 Data required. The following data will be obtained.

a. Control separation and dimensional measurements.

b. Ability of operators to operate the sets while wearing regular, arctic, and NBC gear.

c. Anthropometric and demographic data of test participants.

d. Results of interview and questionnaires administered to test personnel.

e. Results of completed checklists.

f. An assessment of controls and displays.

g. The adequacy of NET.

h. RGL of the operation and maintenance manuals.

i. Photographs and videotapes of HFE problems associated with the setup, operation, or maintenance of the generator sets.

j. Performance times of critical maintenance tasks while maintainers are outfitted in arctic and NBC ensembles.

103.1.4 Results.

a. Qualitative results of observations, checklists and questionnaires will be summarized and presented in narrative and tabular form.

b. The degree to which the generators conform or do not conform to HFE standards and requirements will be presented. Instances of nonconformance will be supported by measurements and photographic illustrations. The causes and consequences of nonconformance will be assessed with regard to the effect on mission performance. Any degradation of the systems man-item relationship with regard to safety will be assessed and corrective action recommended. Personnel performance reliability will be assessed in terms of frequency and consequence of human error committed during preparation, operation, and maintenance of the generators. Subjective data analysis will include a structured interview follow-up of all unfavorable/negative comments to arrive at a description of the cause and possible corrective action.

c. The following will constitute failure of this test:

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(1) Controls, displays, or labeling that do not conform to visual displays, controls, and labeling requirements of MIL-STD-1472 in relation to appearance, spacing, size, or location.

(2) Inability to successfully operate or maintain the sets when personnel are wearing regular, NBC, or MOPP IV gear.

(3) Workspace and maintenance access openings that do not allow personnel, with anthropometrical dimensions between the 5th through the 95th percentile, to perform maintenance.

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TABLE 103.1- II - HFE questionnaire

How would you rate the adequacy of the following?

Rating Scale
- 6 Excellent
- 5 Very Good
- 4 Adequate
- 3 Not Quite Adequate
- 2 Poor
- 1 Extremely Poor

Human Factors Engineering - Adequacy	6	5	4	3	2	1
1. Before, during, and after operation checklist.						
2. Display panels.						
3. Space provided to service generator.						
4. Accessibility of hand controls.						
5. Illumination of instruments during night operation.						
6. Protection of operator from moving parts by guards and warning panels.						
7. Lifting provisions.						
8. Access for using test equipment.						
9. Standard tools and test equipment.						
10. Technical manuals for operations and maintenance.						
11. Based upon the previous questions, rate the OVERALL ADEQUACY of the generator.						

Human Factors Engineering - Tasks	6	5	4	3	2	1
1. Readings warnings or instruction labels.						
2. Connecting and disconnecting power cables.						
3. Operation and maintenance while wearing arctic clothing.						
4. Operation and maintenance while wearing NBC clothing.						
5. Reading and understanding the material presented in the technical manuals.						
6. Set up for operation.						
7. Operation during hours of darkness.						
8. Based upon the previous questions, rate the OVERALL EASE OF OPERATION.						

Please rate how often the following occur?

Rating Scale
- 6 Almost Never
- 5 Very Seldom
- 4 Seldom
- 3 Often
- 2 Very Often
- 1 Almost Always

Human Factors Engineering - Intensity	1	2	3	4	5	6
1. The vibration level during operation.						
2. The noise level during operation.						
3. Exhaust fumes during operation.						

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Human Factors Engineering - Frequency		1	2	3	4	5	6
1.	Requirement for special tools and test equipment.						
2.	Glare on operating instruments and gauges.						

TABLE 103.1- III - HFE checklists

No.	Items	Yes	No	Remarks
	HFE Design - Controls, Displays, and Marking			
1	**Controls**			
a	Are all adjustments located on single panel?			
b	Are controls placed on the panel in the order they will normally be used?			
c	When controls are used in a fixed procedure, are they numbered to indicate?			
d	Are controls labeled with functional statements?			
e	Are control-position markings descriptive rather than coded or numbered?			
f	Are control scales fine enough to permit accurate setting?			
g	Except for detents or selector switches, do the controls have smooth, even resistance to movements?			
h	Are concentric knobs adequately coded to avoid confusion?			
i	Are adjustment controls easy to set and lock?			
j	Do all physical adjustment procedures provide visual, auditory, or tactical feedback?			
k	Are controls free of excessive backlash (play) that could require needless readjustment?			
l	Are primary and emergency controls easily identifiable both visually and nonvisually?			
m	Can controls be operated by personnel wearing arctic and NBC clothing?			
n	The method used to prevent accidental activation of the control, if any does not increase the time required to operate the control to such an extent that it is unacceptable.			
2	**Displays**			
a	When this equipment is placed in ways that it will typically be used, can the display be easily read?			
b	The information presented is necessary for the decisions or actions required of the operator.			
c	The information is presented in the most immediately meaningful form, i.e., no interpretation or decoding is required.			
d	The information is displayed to the accuracy required by the decisions or actions of the operator.			
e	Are display scales limited to only that information needed to make decisions or to take some action?			
f	Information is current, that is, lag is minimized.			
g	Failure is clearly shown or the operator is otherwise warned.			
h	The contrast ratio and illumination of controls and/or displays are sufficient under all expected light conditions.			
i	A warning device is provided to indicate significant deviations from normal operating conditions.			

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Yes - Adequate No - Inadequate NA - Not Applicable

No.	Items	Yes	No	Remarks
3	***Miscellaneous***			
a	Vibration and noise are kept below levels that might impair the efficiency of personnel.			
b	Visibility provides the maximum field of view possible in consonance with station, task requirement, and body conformation.			
c	Illumination of controls and displays is sufficient for the operators to carry out necessary tasks.			
d	Vibrations do not affect operator performance in reading dials and manipulating controls.			
e	No material within the operator's vision is capable of reflecting glare sufficiently to impair vision during day or night operation.			
	Maintainability			
1	***Handles***			
a	When possible, handles are provided on covers, drawers, and components to facilitate handling.			
b	When handles cannot be provided hoist and lift points are clearly marked.			
2	***Covers***			
a	Method of opening a cover is evident from the construction of the cover itself. If not, an instruction plate is permanently attached to the outside of the cover.			
b	Hinges are used where possible to reduce the number of fasteners required.			
c	When a hinged cover is used, a space equal to the swept volume of the cover is provided (e.g., opening of the cover is not obstructed by bulkheads brackets, etc.).			
d	Structural members, other components, etc., do not interfere with removal of a cover.			
e	It is evident when the cover is in place but not secured?			
f	If instructions applying to a covered unit are lettered on , hinged door, the lettering is properly oriented for reading when the door is open.			
g	A minimum number and type of fasteners are used, commensurate with requirements for stress, bonding, etc.			
h	When possible, the same size and type of fasteners are used for all covers, cases, and access doors.			
i	Captive nuts and bolts are used where feasible.			
3	***Location of Replaceable Components***			
a	Large components which are difficult to remove are mounted so that they do not prevent access to other components.			
b	Components are placed to allow sufficient space for use of test equipment and other required tools without difficulty or hazard.			
c	All throwaway components are accessible without removal of other components.			
d	Structural members of the frame do not prevent access to components.			
Yes - Adequate No - Inadequate NA - Not Applicable				

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TABLE 103.1-III - HFE checklists – Continued

No.	Items	Yes	No	Remarks
e	Delicate components are so located or guarded that they will not be damaged while the unit is being handled or worked on.			
f	Sensitive adjustments are so located or guarded that they cannot be accidentally disturbed.			
g	Internal controls are located at a safe distance from dangerous voltages or access to dangerous voltages is prevented by suitable barriers.			
4	***Conductors and Cables***			
a	Conductors are bound into cables and held by means of lacing twine or other acceptable means.			
b	Long conductors or cables, internal to equipment, are secured to the chassis by cable clamp.			
c	Cables are long enough so that each functioning component can be checked in a convenient place or, if this is not feasible, extension cables/devices are provided.			
d	Cables are long enough to permit jockeying or movement of components when it is difficult to connect or disconnect other cables.			
e	Electrical cables are not to be routed below fluid lines.			
f	Cables are routed so they cannot be walked on or used for handholds.			
g	Cables are easily accessible for inspection and repair.			
h	Cables are so routed that they need not be bent or twisted sharply or repeatedly.			
i	Input and output cables, with the exception of test cables, do not terminate on a control-display panel.			
j	If test cables terminate on control-display panels test receptacles are located so that their associated cables do not interfere with controls and displays.			
5	***Connectors***			
a	One-turn or other quick- disconnect plugs are used.			
b	When dirt and moisture are a problem, plugs have an attached cover.			
c	Connectors are located far enough apart so that they can be grasped firmly for connection and disconnection.			
d	Plugs are designed so that it is impossible to insert the wrong plug in a receptacle.			
e	Socket rather than plug contacts are "hot".			
f	Test points to determine that a unit is malfunctioning are provided.			
g	Appropriate test provided when a component is not completely self-checking.			
6	***Fuses and Circuit Breakers***			
a	Fuses and circuit breakers are so located that they can be easily seen and quickly replaced or reactivated by personnel wearing clothing appropriate to environment of interest.			
b	No special tools are required for fuse replacement.			
Yes - Adequate No - Inadequate NA - Not Applicable				

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TABLE 103.1-III - HFE checklists – Continued

No.	Items	Yes	No	Remarks
7	***Tools***			
a	Variety of tools is held to a minimum.			
b	As few special tools as possible are required.			
c	Tools to be used near high voltage are adequately insulated.			
d	Metal handles are avoided on tools likely to be used in extreme cold or heat.			
8	***Lubrication***			
a	Equipment containing mechanical components either has provision for lubrication without disassembly or does not require lubrication.			
b	When lubrication is required, the type of lubricant to be used and the frequency of lubrication is specified by a label at or near the lubrication point.			
Yes - Adequate No - Inadequate NA - Not Applicable				

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APPENDIX B

PACKAGING REQUIREMENTS

1 First article pack. Unless otherwise specified, the contractor shall furnish a first article pack for each item for examination within the time frame specified, to prove prior to the start of production packaging that the applied preservation, packing and marking comply with the requirements of this specification. Examination shall be as specified in paragraph 6 and shall be subject to surveillance and approval by the government. The first article packs may utilize either the first article models or production models. When the first article models are utilized, and the government requests a comparison between the first article models and the production models, any preservation and packing shall be removed at no expense to the government.

2 General. All connector protective caps and covers shall be secured in position on their companion connectors. Contact preservatives shall not be used on connectors or connector components.

3 Preservation. Preservation shall be level A or C as specified.

3.1 Level A.

3.1.1 Electrical distribution equipment end item.

3.1.1.1 Cable assemblies and cable assembly extensions. Each cable assembly and cable assembly extension shall have their ends protected with barrier material conforming to MIL-PRF-121, Type II, Class 2, held in place with no less than one complete wrap of tape conforming to ASTM D5486-93, Type IV. The wrap shall cover the end and extend back over the side a minimum of 2 inches beyond the connector or pigtail end. Additional tape shall be applied over the barrier material at the pigtail end to prevent unsheathed wire ends from spreading apart. Each cable shall then be coiled to a compact diameter, sufficient to prevent excessive bending of the cable. Each coil shall be tied or secured a minimum of three places equidistantly spaced around the circumference of the coil.

3.1.1.2 Loose receptacles. Each loose receptacle shall be preserved IAW MIL-STD-2073-1E, Method 32.

3.1.1.3 Loose connectors. Each loose connector shall be preserved IAW MIL-STD-2073-1E, Method 32.

3.1.1.4 Rope assemblies. Each rope assembly shall be coiled and secured in a tight, compact coil and then preserved IAW MIL-STD-2073-1E, Method 31.

3.1.1.5 Branch circuit assemblies. The connectors at the ends of the branches shall be protected with barrier material as specified in 3.1.1.1. The receptacles shall be protected in the same manner. Each circuit assembly shall then be coiled in the manner specified in 3.1.1.1 with additional ties as necessary to prevent movement of the receptacles.

3.1.1.6 Distribution centers. All protective cover anchoring chains shall be taped to the connector with tape conforming to ASTM D5486-93, Type IV. The distribution center lid shall be secured in the closed position with the latches provided. The center shall then be covered with a wrap or bag constructed of plastic film IAW A-A-3174, Type I, Class 1, with a minimum thickness of 0.004 inches. The wrap or bag shall be of such dimension to completely cover the top, sides and ends of the center with enough surpluses to be tucked under the body overlapping

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itself a minimum of 3 inches. The film shall be taped in place at the overlap with tape conforming to ASTM D5486-93, Type IV. The lower portion of the support frame shall not be covered with the film.

3.1.2 Consolidation.

3.1.2.1 M46. All electrical assembly components for the M46, electrical kit, which have been preserved as specified in 3.1.1.1 through 3.1.1.5 as applicable, in the quantities specified on TA13226E7046, shall be placed in the assigned utility storage container. Cushioning material conforming to A-A-1898, Grade 2, style and size optional, 1/4 inch thick, shall be utilized within the container to cushion the components from each other and to prevent movement. The container shall be secured in the closed position by the means provided.

3.1.2.2 Distribution systems.

3.1.2.2.1 M60. All components for M60, single-phase, 60 ampere system, except the distribution center, which have been preserved as specified in 3.1.1.1 through 3.1.1.5as applicable, in the quantities specified on TA13229E6352, shall be placed in the assigned storage and transit container. Cushioning and container closure shall be as specified in3.1.2.1.

3.1.2.2.2 M40. All components for M40, three-phase, 40 ampere system, except the distribution center, which have been preserved as specified in 3.1.1.1 through 3.1.1.5applicable, in the quantities specified on TA13229E6353, shall be placed in the assigned storage and transit container. Cushioning and container closure shall be as specified in 3.1.2.1.

3.1.3 Electrical distribution equipment auxiliary components.

3.1.3.1 Cable assemblies. Cable assemblies shall be preserved as specified in 3.1.1.1. The coiled cable shall be placed in a bag and sealed IAW MIL-STD-2073-1E, Method 31.

3.1.3.2 Rope assemblies. Rope assemblies shall be preserved as specified in 3.1.1.4.

3.1.3.3 Branch circuit assemblies. Branch circuit assemblies shall be preserved as specified in .3.1.1.5. The coiled circuit shall be placed in a bag and sealed IAW MIL-STD-2073-1E, Method 31.

3.1.3.4 Intermediate containers. The individual auxiliary components which have been preserved as specified in 3.1.3.1 thru 3.1.3.3 shall be placed in intermediate containers. The intermediate containers shall be close-fitting boxes conforming to ASTM D5118/D5118 and ASTM D1974/D1974M, type CF, class domestic, variety SW, grade 2000 and style optional. Box closure shall be as specified in the appendix to the box specification. Like components only shall be placed in an intermediate container. Unless otherwise specified, the quantity of a component within an intermediate container shall be sufficient to result in a gross weight of not less than 45 pounds but not more than 50 pounds.

3.1.3.5 Universal adapter. The universal adapter shall be preserved as specified in 3.1.1.6 for the distribution center. Additionally, each adapter shall be placed in a unit container complying with the requirements of ASTM D5118/D5118 and ASTM D1974/D1974M, Type CF, Grade V3C, style optional. Cushioning and blocking shall be applied to protect the adapter

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and to prevent free movement within the container. Container closure shall be IAW Method V of the appendix to ASTM D1974/D1974M.

3.2 Level C. All components included in the electrical distribution equipment shall be preserved in a manner to prevent deterioration and damage during storage and shipment from the contractor to the initial destination.

4 Packing. Packing shall be level A, B, or C as specified.

4.1 Level A.

4.1.1 Electrical distribution equipment end items.

4.1.1.1 Electrical assembly. Each electrical assembly (see TA13226E7046), preserved and placed in the utility storage container as specified in 3.1.2.1, shall be individually packed in a close-fitting box conforming to ASTM D6251/D6251M, Class 2, Overseas Boxes. Box closure shall be IAW the appendix thereto and the strapping shall be IAW ASTM D3953, Type I or 2, zinc-coated, size as applicable and ASTM D4675.

4.1.1.2 Distribution systems.

4.1.1.2.1 M60. All components for each M60 system (see TA13229E6352), shall be packed together in a box IAW Drawing 13230E7021 or 08-26001. The distribution center, preserved as specified in 3.1.1.6, shall be secured to the bottom of the box to prevent the center from moving within the box. Skids or intermediate cleats shall be added to the bottom of the box to enable the center to be properly secured. The box shall be constructed with one wood partition which divides the box vertically into two cells. One cell shall be for the center and the other cell shall be for the balance of the system's components which are enclosed within the applicable storage and transit container. The inside dimensions of the box shall provide a minimum of 1 inch clearance between any interior box surface and the top, sides and ends of the center, which is secured on one side of the partition, while providing adequate space on the other side of the partition for the filled storage and transit container. The partition shall provide the necessary rigidity, through its material strength and its method of attachment to the box body, to prevent movement of the filled storage and transit container toward, the distribution center. Blocking shall be provided, as necessary, to prevent movement of the filled storage and transit container toward the distribution center. Blocking shall be provided, as necessary, to prevent movement of the filled storage and transit container in other directions. Box closure and strapping shall be as specified in 4.1.1.1.

4.1.1.2.2 M40. All components for each M40 system (see TA13229E6353), shall be packed together in a box IAW DWG with 13230E7021 or 08-26000 and as specified in 4.1.1.2.1.

4.1.1.3 Feeder system.

4.1.1.3.1 M100. All components for each M100 system (see TA13229E6351), shall be packed together in a box IAW Drawing with 13230E7020 or 08-26002. The feeder system, preserved as specified in 3, shall be secured to the bottom of the box at one end as specified in 4.1.1.2.1. Additional wood skids shall be added to the bottom of the box to enable the center to

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be properly secured. The box shall be constructed with one wood partition which divides the box vertically into two cells. One cell shall be for the center and the other cell shall be for the three coiled cables making up the balance of the system's components. One inch clearance shall be provided for the center as specified in 4.1.1.2.1 and the partition shall provide the necessary rigidity to prevent movement of the coils toward the distribution center. The coils shall be cushioned with cushioning material specified in 3.1.2.1 and blocking shall be provided, as necessary, to prevent movement of the coils in other directions. Box closure and strapping shall be as specified in 4.1.1.1.

4.1.1.3.2 M200. All components for each M200 system shall be packed shall be packed together in a box IAW DWG with 13230E7021 or 08-26003 and as specified in 4.1.1.2.1 as specified in 4.1.1.3.1 for the 100 ampere system except that the referenced drawing shall be TA13229E6350 and that five coiled cables are required.

4.1.1.4 Electrical distribution equipment auxiliary components. The electrical distribution equipment auxiliary components which have been preserved as specified in 3 and the 13229E6362, utility container, shall be packed in close-fitting boxes conforming to ASTM D6251/D6251M, Class 2, Overseas Boxes, optional. Box closure shall be as specified in the box specification and the appendix thereto and strapping shall be IAW ASTM D 3953, Type 1 or 2, zinc-coated, size as applicable and ASTM D 4675. Like components only shall be packed in each box.

4.2 Level B. Packing shall be as specified for level A except any boxes sh

4.3 all be domestic type and strapping may be coated finish.

4.4 Level C. All of the electrical distribution equipment shall be packed to ensure carrier acceptance and safe delivery to destination at lowest ratings IAW Uniform Freight Classification Rules or National Motor Freight Classification Rules. Only like components shall be packed in a shipping container for the auxiliary components.

5 Marking. In addition to any special or identification markings required by the contract or purchase order, each container shall be marked IAW with MIL-STD-129. Center of balance markings shall be applied as specified therein.

6 INSPECTION OF PACKAGING

6.1 First article packaging inspection. The first article packages examined for the defects 1isted in paragraph 6.1.1.3, Table I. The presence of one or more shall be cause for rejecting the first article pack.

6.1.1 Quality conformance inspection of packaging.

6.1.1.1 Electrical distribution equipment end item. For the purpose of inspection only, an end item unit of product shall consist of any complete end item covered by this specification that has been prepared for shipment.

6.1.1.2 Sampling. Sampling for examination shall be selected IAW MIL-STD-1916.

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6.1.1.3 Examination. Samples selected IAW MIL-STD 916 shall be examined for the major defects 1isted in Table I. Presence of one or more defects shall be cause for rejection.

TABLE I– Packaging examination schedule (requirements).

No	Defect	Requirement Paragraph		
		Level A	Level B	Level C
P1	Protective caps and covers not installed as specified.	2		
P2	Contact preservative used where not permitted.	2		
PRESERVATION **End Item**				
P3	Ends of cables not protected as specified.	3.1.1.1		
P4	Cables not coiled as specified.	3.1.1.1		
P5	Loose receptacles not preserved as specified.	3.1.1.2		
P6	Connectors not preserved as specified.	3.1.1.3		
P7	Rope assemblies not coiled and preserved as specifies.	3.1.1.4		
P8	Branch circuit assembly end connectors and receptacles not protected as specified.	3.1.1.5		
P9	Branch circuit assembly not coiled as specified.	3.1.1.5		
P10	Protective cover anchoring chains not taped to connector as specified for the distribution center.	3.1.1.6		
P11	Distribution center 1id not secured in closed position as specified.	3.1.1.6		
P12	Distribution center not covered with wrap or bag as specified.	3.1.1.6		
P13	Wrap or bag not taped in place on distribution center as specified.	3.1.1.6		
P14	Electrical assembly components not consolidated, cushioned and secured in the assigned storage and transit container as specified.	3.1.2.1		

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TABLE I– Packaging examination schedule (requirements).

No	Defect	Requirement Paragraph		
		Level A	Level B	Level C
P15	Single phase, 60 ampere distribution system components, excluding the distribution center not consolidated, cushioned and secured in the assigned storage and transit container as specified.	3.1.2.2.1		
P16	Three phase, 40 ampere distribution system components, excluding the distribution center, not consolidated, cushioned and secured in the assigned storage and transit container as specified.	3.1.2.2.2		
Auxiliary Components				
P17	Cable assemblies not preserved as specified.	3.1.3.1		
P18	Rope assemblies not preserved as specified.	3.1.3.2		
P19	Branch circuit assemblies not preserved as specified.	3.1.3.3		
P20	Intermediate containers not as specified.	3.1.3.4		
P21	Applicable components not placed in intermediate containers as specified.	3.1.3.4		
P22	Unlike components placed in same intermediate container.	3.1.3.4		
P23	Gross weight of intermediate container not as specified.	3.1.3.4		
P24	Universal adapter not preserved and placed in container connector as specified for the distribution center.	3.1.3.5		
P25	Components not preserved as specified for level C preservation.			3.2
PACKING **End Items**				
M60 Distribution system				
P26	Electrical assembly not packed as specified	4.1.1.1	4.2	

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TABLE I– Packaging examination schedule (requirements).

No	Defect	Requirement Paragraph		
		Level A	Level B	Level C
P27	Shipping container not as specified.	4.1.1.2.1	4.2	
P28	Distribution centers not secured to the shipping container bottoms as specified.	4.1.1.2.1	4.2	
P29	Vertical wood partition not installed in shipping containers as specified.	4.1.1.2.1	4.2	
P30	Clearance around distribution centers not as specified	4.1.1.2.1	4.2	
P31	Filled storage and transit container not blocked within its cell as required.	4.1.1.2.1	4.2	
P32	Shipping container closure and strapping not as required.	4.1.1.2.1	4.2	
M40 Distribution system				
P33	M40 system not packed as specified	4.1.1.2.2	4.2	
P34	Shipping container not as specified.	4.1.1.2.2	4.2	
P35	Distribution centers not secured to the shipping container bottoms as specified.	4.1.1.2.2	4.2	
P36	Vertical wood partition not installed in shipping containers as specified.	4.1.1.2.2	4.2	
P37	Clearance around distribution centers not as specified	4.1.1.2.2	4.2	
P38	Filled storage and transit container not blocked within its cell as required.	4.1.1.2.2	4.2	
P39	Shipping container closure and strapping not as required.	4.1.1.2.2	4.2	
M100 Feeder systems				
P40	Shipping containers for the feeder systems not as specified.	4.1.1.3.1	4.2	

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TABLE I– Packaging examination schedule (requirements).

No	Defect	Requirement Paragraph		
		Level A	Level B	Level C
P41	Feeder systems not secured to shipping container bottoms as specified.	4.1.1.3.1	4.2	
P42	Vertical wood partition not installed in shipping containers as specified.	4.1.1.3.1	4.2	
P43	Clearance around feeder systems not as specified.	4.1.1.3.1	4.2	
P44	Coils not cushioned and blocked within their cell as specified.	4.1.1.3.1	4.2	
P45	Shipping container closure and strapping not as specified.	4.1.1.3.1	4.2	
M200 Feeder systems				
P46	Shipping containers for the feeder systems not as specified.	4.1.1.3.2	4.2	
P47	Feeder systems not secured to shipping container bottoms as specified.	4.1.1.3.2	4.2	
P48	Vertical wood partition not installed in shipping containers as specified.	4.1.1.3.2	4.2	
P49	Clearance around feeder systems not as specified.	4.1.1.3.2	4.2	
P50	Coils not cushioned and blocked within their cell as specified.	4.1.1.3.2	4.2	
P51	Shipping container closure and strapping not as specified.	4.1.1.3.2	4.2	
Auxiliary components				
P52	Auxiliary components not packed as specified.	4.1.1.4	4.2	
P53	Shipping container not as specified	4.1.1.4		

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APPENDIX B

PACKAGING REQUIREMENTS

TABLE I– Packaging examination schedule (requirements).

No	Defect	Requirement Paragraph		
		Level A	Level B	Level C
P54	Box closure and strapping not as specified.	4.1.1.4		
P55	Unlike components mixed in shipping container	4.1.1.4		4.3
P56	Components not packaged as specified for level C.			4.3
P57	Marking illegible, missing, or incomplete	5.5	5.5	5

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CONCLUDING MATERIAL

Custodian:

Army - CR4

Preparing Activity:

Army - CR4
(Project 6150-2012-002)

Review Activities:

DLA – GS

Note: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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